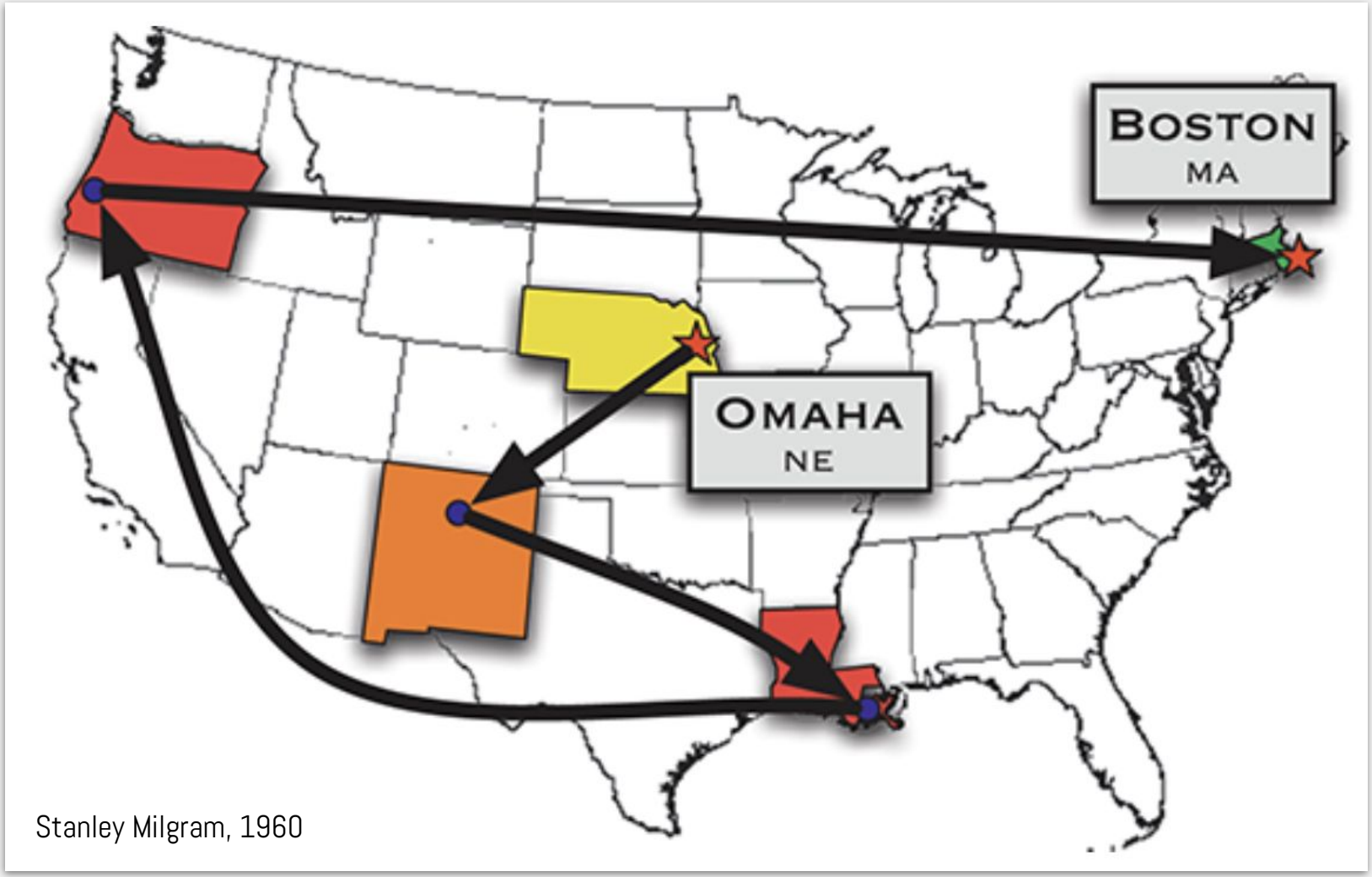


Small World

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UFRN UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE

DCA



Stanley Milgram, 1960

Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

COVID-19: A scholarly production dataset report for research analysis

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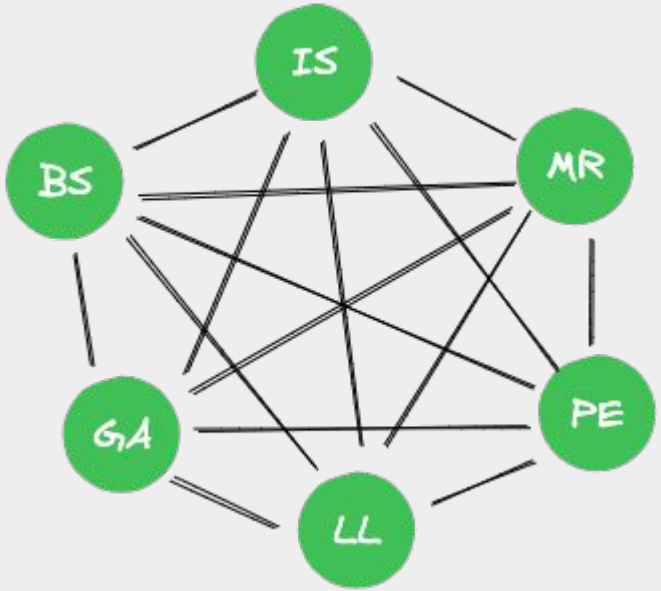
Keywords:
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SARS-CoV-2
Pandemic
Data Science
Bibliometrics
Scientometrics

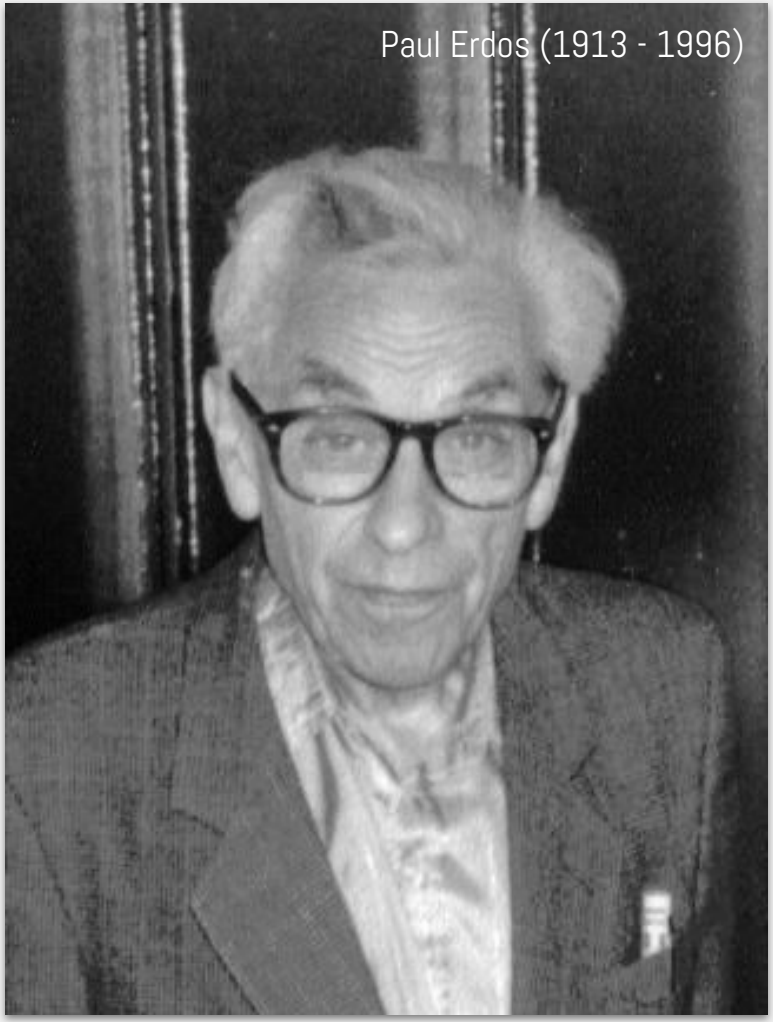
ABSTRACT

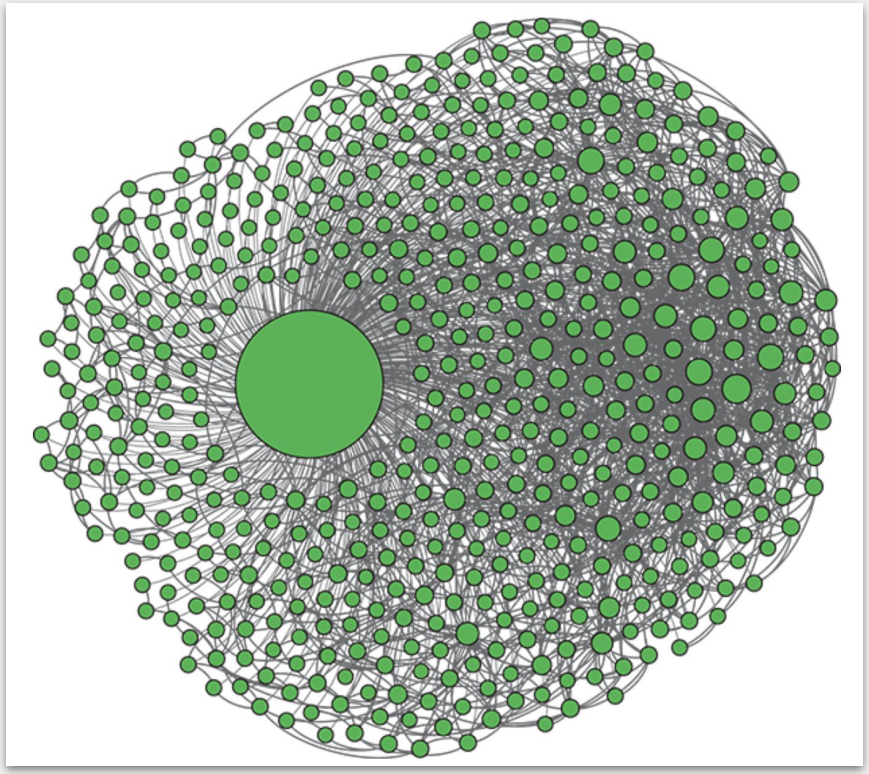
COVID-2019 has been recognized as a global threat, and several studies are being conducted in order to contribute to the fight and prevention of this pandemic. This work presents a scholarly production dataset focused on COVID-19, providing an overview of scientific research activities, making it possible to identify countries, scientists and research groups most active in this task force to combat the coronavirus disease. The dataset is composed of 40,212 records of articles' metadata collected from Scopus, PubMed, arXiv and bioRxiv databases from January 2019 to July 2020. Those data were extracted by using the techniques of Python Web Scraping and preprocessed with Pandas Data Wrangling. In addition, the pipeline to preprocess and generate the dataset are versioned with the Data Version Control tool (DVC) and are thus easily reproducible and auditable.

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Coauthorship networks



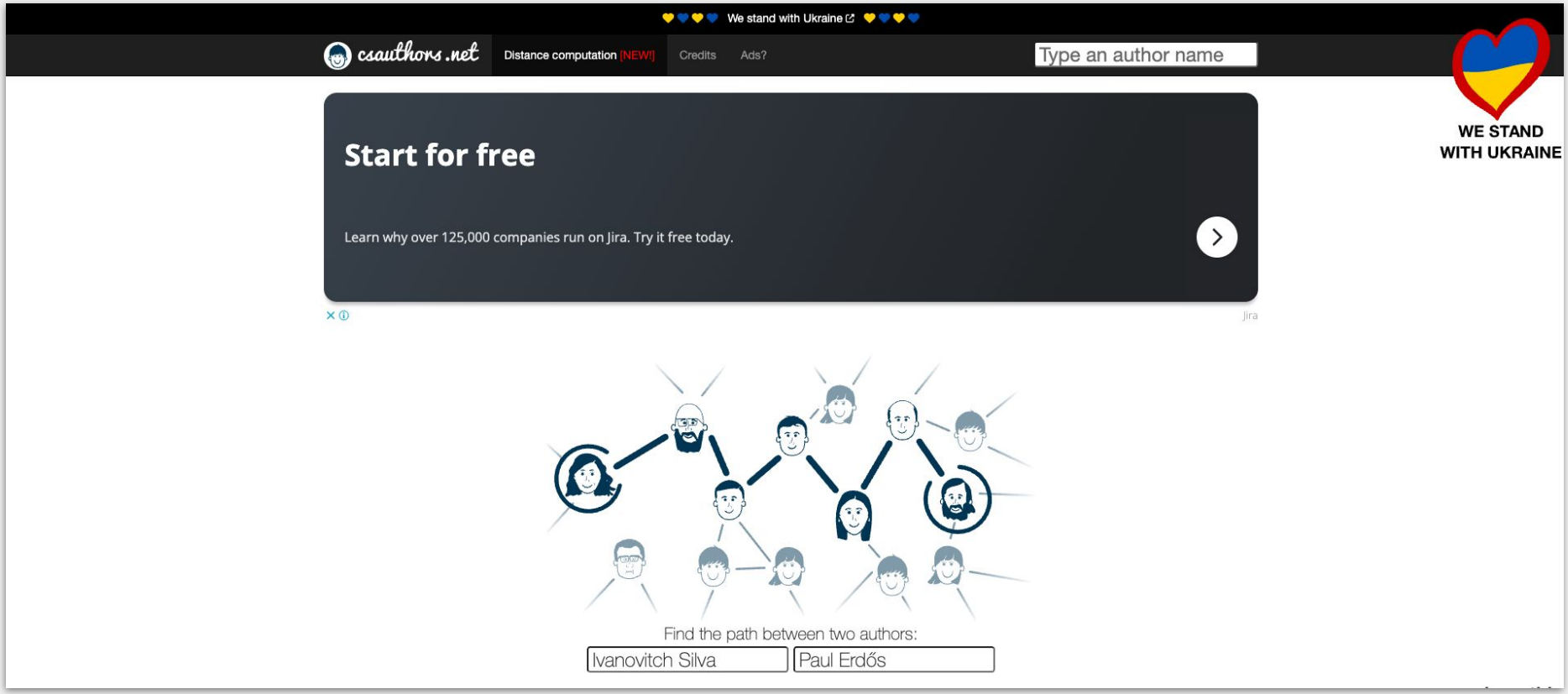


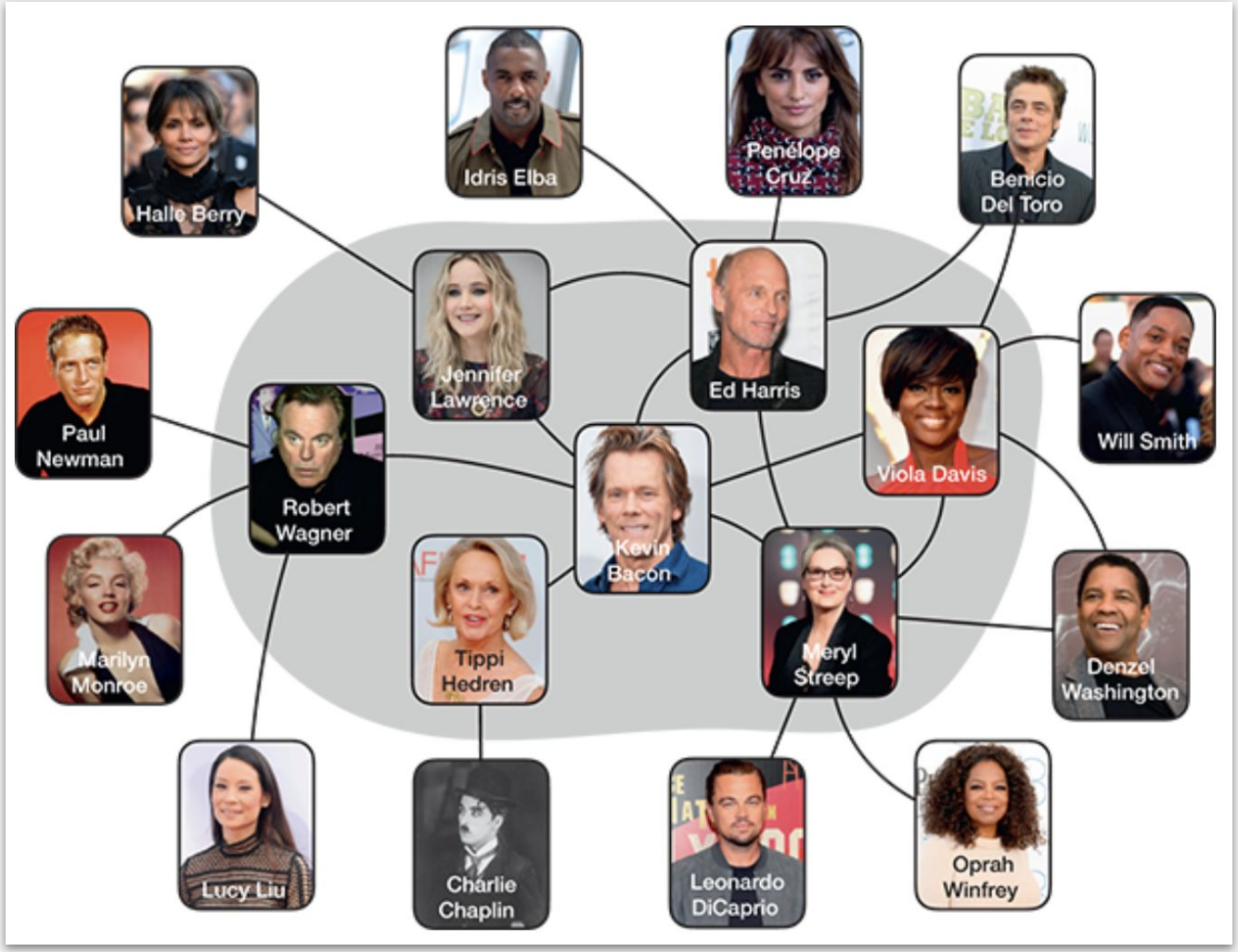


Ego Network

Paul Erdős was a famous mathematician who made critical contributions to network science. Mathematicians are fond of studying their distance in the coauthorship network from the particular node corresponding to Erdős. They call this distance their **Erdős number** .

https://www.csauthors.net/distance/ivanovitch-silva/paul-erdos





The Oracle of Bacon

http://oracleofbacon.org/

The Wiki Game App
Addictive solo-play fun! Featuring 5 game modes with 200 unique levels. *Now includes minimal path unlocking*!
GET THE IPHONE APP! TEXT ME THE APP!

CURRENT ROUND



START
Walmart



GOAL
Hebrew language

PLAY NOW!
21s

ROUND RESULTS | DAY LEADERS | WEEK LEADERS | ALL-TIME LEADERS

YOUR WINS ARE NOT SAVED! CREATE ACCOUNT! | LOGIN

Boy Scouts of America → Sodium chloride
(4 minutes ago) ← →

#1.**TealDeer46**(2000pts)

WIN #1
Boy Scouts of America → New Mexico → Mining → Sodium chloride
(800PTS · 4CLICKS · 19SECS)

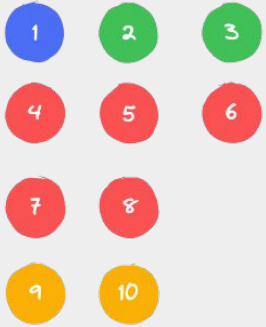
WIN #2
Boy Scouts of America → New Mexico → Economy of New Mexico → Manganese → Chlorine → Sodium chloride
(1200PTS · 6CLICKS · 82SECS)

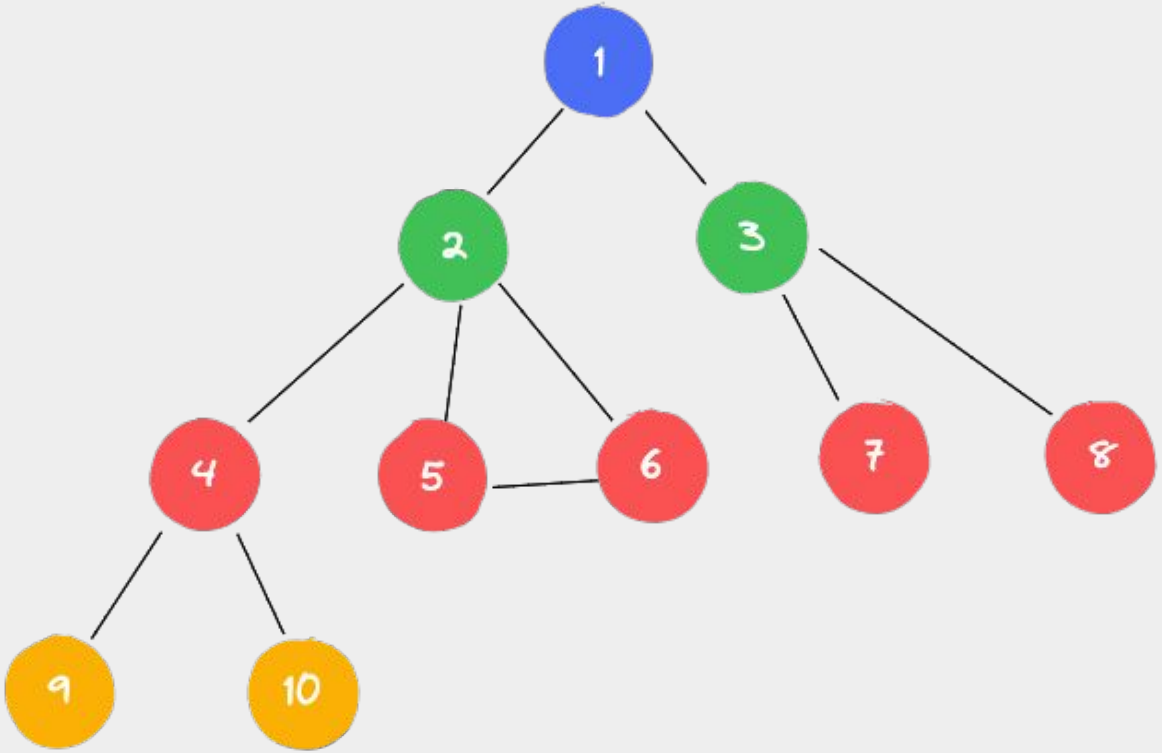
#2.**CrimsonChinchilla37**(600pts)

Feedback?

Breadth-First Search (BFS) vs Depth-First Search (DFS)

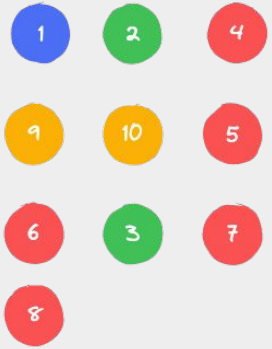
Visited Nodes

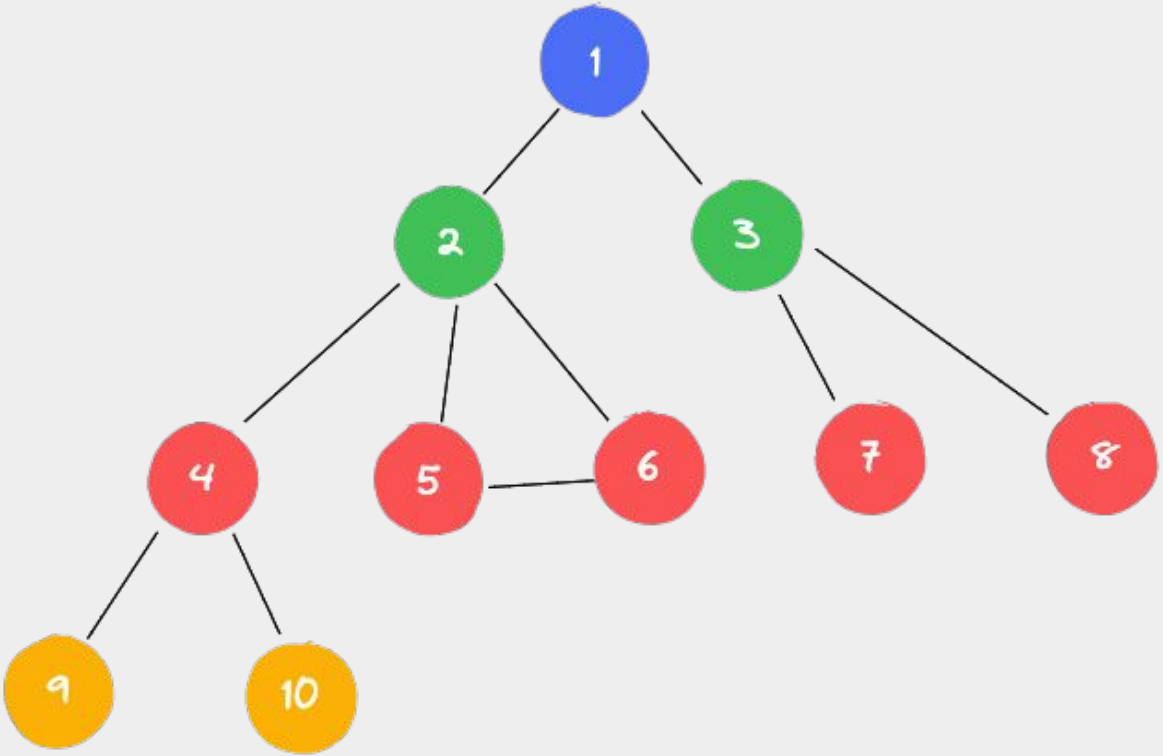


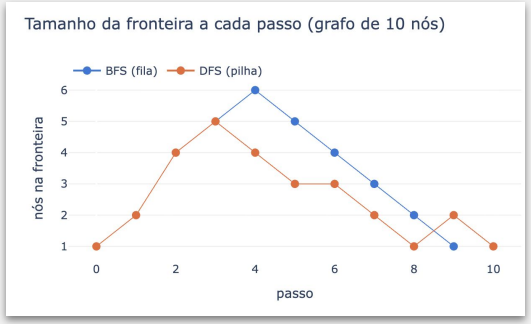


Breadth-First Search (BFS) vs Depth-First Search (DFS)

Visited Nodes



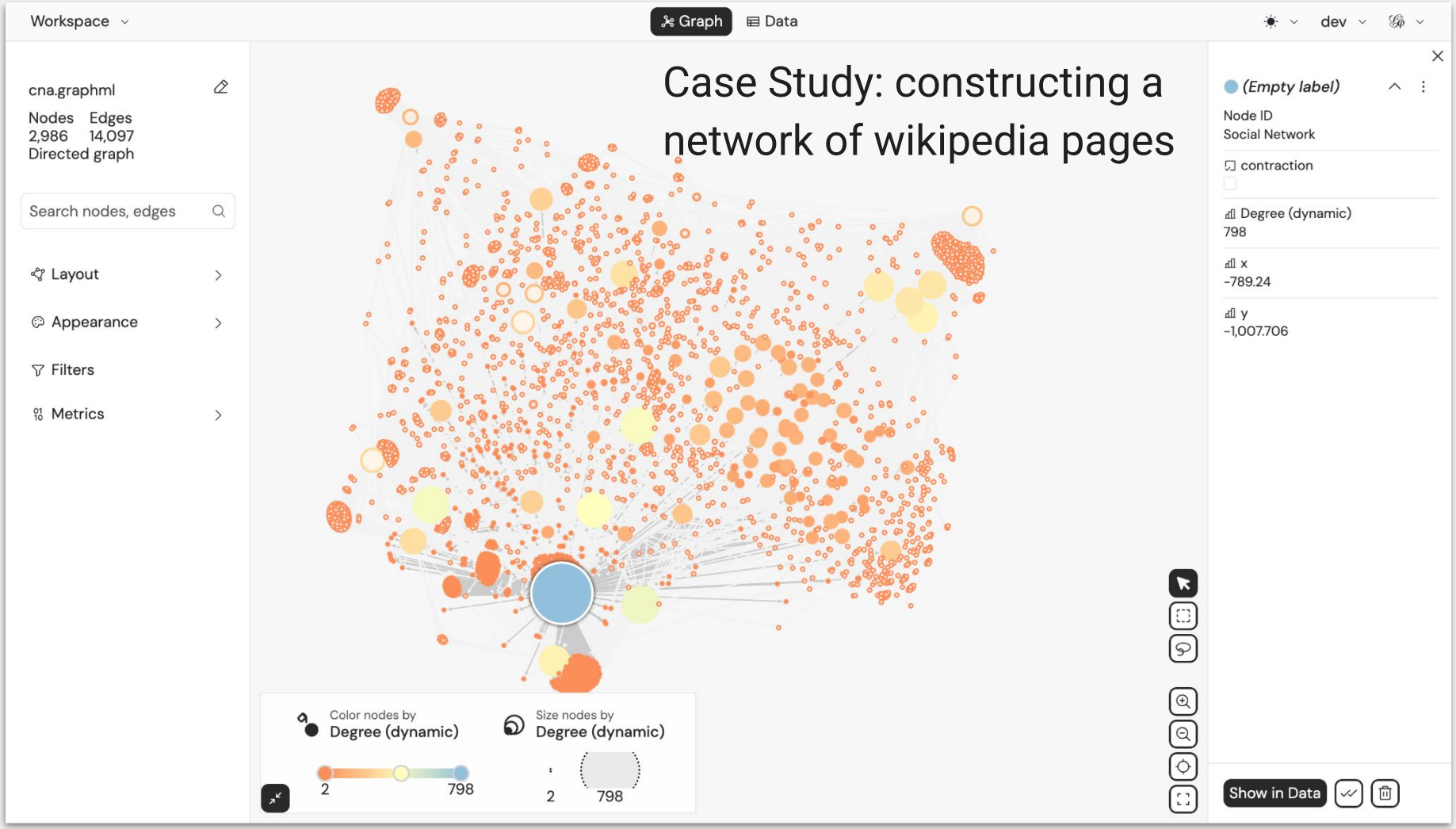




BFS vs DFS

Graph Transversal & WikiGame

	Breadth-First Search (BFS)	Depth-First Search (DFS)
Exploration strategy	Level-by-level (layered expansion)	Goes deep along one path before backtracking
Data structure	Queue (FIFO)	Stack / Recursion (LIFO)
Goal guarantee	Finds shortest path	Does not guarantee shortest path
Time complexity (algorithm)	O(V + E)	O(V + E)
Space complexity (algorithm)	O(V) (stores frontier)	O(h) (depth of search, worst-case O(V))
Branching factor (d)	Expands all neighbors at each level	Expands only one branch at a time
Frontier growth	O(d^k) (exponential in depth k)	O(d · k) (linear in depth)
Memory usage	High (stores full layer)	Low (stores path only)
Risk	Memory explosion	Deep/unproductive paths
Best use cases	Shortest path, minimum hops	Exploration, large/deep graphs


Workspace
Graph
Data
dev
cna.graphml
Nodes 2,986
Edges 14,097
Directed graph
Search nodes, edges
Layout
Appearance
Filters
Metrics
Case Study: constructing a network of wikipedia pages
Color nodes by
Degree (dynamic)
2
798
Size nodes by
Degree (dynamic)
2
798
(Empty label)
Node ID
Social Network
contraction
Degree (dynamic)
798
x
-789.24
y
-1,007.706
Show in Data